

UNIT-2: LASERS & OPTICAL FIBERS

LASERS

Exercise Problems:

1. A pulsed laser emits photons of wavelength 780 nm with 20 mW average power/pulse. Calculate the number of photons contained in each pulse if the pulse duration is 10 ns.
2. A laser operating at 632.8 nm emits 3.182×10^{16} photons per second. Calculate the output power of the laser if the input power is 100 W. Also, find the percentage power converted into coherent light energy.
3. A laser medium at thermal equilibrium temperature 300 K has two energy levels with a wavelength separation of 1 μm . Find the ratio of population densities of the upper and lower levels.
4. The ratio of population of two energy levels is 1.059×10^{-30} . Find the wavelength of light emitted at 330 K.
5. Calculate the ratio of (i) Einstein coefficients (ii) Stimulated to spontaneous emissions for a system in thermal equilibrium at 300 K in which radiations of wavelength 1.39 μm are emitted.
6. In a particular laser system, $E_{\text{ex}} = 20.66 \text{ eV}$, $E_{\text{meta}} = 18.70 \text{ eV}$ and $E_{\text{gr}} = 0 \text{ eV}$. Find the wavelength emitted and population of metastable level with respect to the excited level at 300 K.
7. The transition to the ground state from the upper and lower energy states in a laser results in emission of photons of wavelengths, 6928 Å. and 6943 Å. respectively. Estimate the energy values of the two energy levels in eV and also their ratio of populations.
8. A laser beam can be focused on an area equal to the square of its λ ($\lambda = 6328 \text{ Å}$). If laser radiates energy at the rate of 1mW, find out the intensity of focused beam.
9. Calculate the divergence of light beam issuing out of laser, which produces spot diameters of 4mm and 6mm at 1mm and 2mm distances respectively.

Assignment Problems:

10. The average output power of laser source emitting a laser beam of wavelength 6328 Å is 5 mW. Find the number of photons emitted per second by the laser source.
11. Find the ratio of population of the two energy states of the active medium producing laser

transition between which has wavelength $6328 \text{ \AA}$. Assume temperature = 300K .

12. A semiconductor laser has a peak emissions radiation of wavelength of $1.24 \text{ }\mu\text{m}$. What is its band gap value in eV?
13. A pulse from laser with power 1 mW lasts for 10 nS . If the number of photons emitted per second is 3.491×10^7 , calculate the wavelength of laser.
14. Calculate the wavelength of emitted radiation from GaAs which has a band gap of 1.44 eV .
15. A typical laser system emits laser beam of wavelength $3.124 \text{ }\mu\text{m}$. What is the energy difference between excited state and metastable state and energy of the emitted photon? If one mole of photons is emitted per second calculate the output power.
16. Find the intensity of a laser of 314mW power and having a diameter of 2mm .
17. A three-level laser emits laser light $5500 \text{ \AA}$. In the absence of pumping (a) what will be the equilibrium ratio of the population of the upper level to that of lower level at 300 K ? (b) At what temperature the above ratio would be $\frac{1}{2}$.

OPTICAL FIBERS

Exercise Problems:

1. Fractional index change of optical fiber and refractive index of core are 0.00515 and 1.533 respectively. Determine the cladding refractive index and critical angle of incidence.
2. An optical fiber has a core material with refractive index 1.55 and its cladding material has a refractive index of 1.50 . The light is launched into it in air. Calculate its numerical aperture, the acceptance angle and also the fractional index change.
3. An optic glass fiber of refractive index 1.45 is to be clad with another glass to ensure internal reflection that will contain light travelling within 5° of the fiber axis. What maximum index of refraction is allowed for the cladding.
4. The N.A of an optical fiber is 0.2 when surrounded by air. Determine the refractive index of its core given the refractive index of the cladding is 1.59 . Also find the acceptance angle when the fiber is in water. Assume the refractive index of water as 1.33 .
5. Calculate the V-number for a fiber of core diameter $40 \text{ }\mu\text{m}$ and with refractive indices of 1.55 and 1.50 respectively for core and cladding when the wavelength of the propagating wave is 1400 nm . Also calculate the number of modes that the fiber can support for propagation. Assume that the fiber is in air.

6. Find the attenuation in an optical fiber of length 500 m, when a light signal of power 100 mW emerges out of the fiber with a power 90 mW.
7. An optical signal loses 85% of its power after traversing a fiber length 400 m. What is the fiber loss?
8. The attenuation in an optical fiber is given as 5 dB/Km, what would be the remaining intensity of light after travelling 5 Km?

Assignment Problems:

9. The refractive indices of core and cladding are 1.50 and 1.48 respectively in an optical fiber. Find the numerical aperture and angle of acceptance.
10. The angle of acceptance of an optical fiber is 30° when kept in air. Find the angle of acceptance when it is in a medium of refractive index 1.33.
11. The attenuation of light in an optical-fiber is 3.6 dB/km. What fraction of its initial intensity remains after (i) 1 km (ii) after 3 km?
12. Calculate the number of modes that can propagate inside an optical fiber, given $n_{\text{core}} = 1.53$, $n_{\text{clad}} = 1.50$, core radius = $50\mu\text{m}$, $\lambda = 1\mu\text{m}$.
13. A fiber 500 m long has an input power of 8.6 mW and output power 7.5 mW. What is the loss specification in cable?
14. What is the index of refraction in a medium where the speed of light is 1.5×10^8 m/s?